Long Branch Remover

Formulas and decision rules as implemented

A reference for what the code actually computes. Every formula and every condition below is what is in `long_branch_rules.jl`, in the order the code applies it. The reasoning behind the choices is in `long_branch_python_vs_julia.pdf`; how to run it is in `HOWTO_run.md`.

> ***Preliminary.*** *This document was produced with Claude AI during the debugging and revision of the Julia implementation, which was itself carried out with Claude AI. It describes code that is still being tested, and should be treated as a working record rather than a settled specification. Anything here may change as testing continues.*

0. Notation

Symbol	Meaning
`K`	trigger value for a rule, from the `-t` file
`q`	safeguard quantile for a rule, from the `-s` file
`m_u`	median of the observations belonging to unit u (a taxon, a clade, or a tree)
`M`	global constant for that rule: the median of all the per-unit medians
`Q(q)`	the q-th quantile of the pooled list of every observation for that rule
`bl`	the branch length being tested

Constant	Value	Meaning
`MIN_CROSS_GENE_OBS`	`3`	trees in which a unit must be observed before it can be tested
`MIN_COMPARATOR_BRANCH`	`1e-6`	absolute backstop: no branch at or below this can serve as a quartet comparator. A branch at the inference floor is unestimated, not short
`COMPARATOR_SCALE_FRACTION`	`0.01`	the working cutoff is this fraction of the dataset's typical terminal branch

The two combine in one helper, evaluated once per run and passed to every quartet function as the keyword `min_comparator_branch`:

```
comparator floor = max( MIN_COMPARATOR_BRANCH , 0.01 * median(all terminal branches) )
```

> `comparator_branch_threshold(global_terminal_bls)`. *The median here is the median of the per-taxon medians, the same M the terminal rule uses, so the floor scales with the dataset rather than with any one gene.*
>
> ***Two distributions, used for different things.*** *The per-unit medians give the trigger; the pooled list of every observation gives the safeguard. A unit votes once in M regardless of how often it was sampled, whereas Q(q) is a percentile of real branches and so counts every observation.*

1. The common trigger

Three of the four rules share one form:

```
trigger_u = (K - 1) * m_u + M
```

When a unit is exactly typical (m_u equals M) this reduces to K * M, so K reads as "how many times the typical value". Both the trigger and the safeguard must be exceeded before anything is nominated:

```
remove  if   bl > trigger_u   AND   bl > Q(q)
```

Both comparisons are strict, so a value sitting exactly on a threshold is retained.

Rule	unit u	m_u is the median of	M is the median of
Terminal	taxon	that taxon's terminal branch across trees	the per-taxon medians

Rule	unit u	m_u is the median of	M is the median of
Clade stem	named clade	that clade's stem across trees	the per-clade medians
Internal	gene tree	all internal branches in that tree	the per-tree medians

2. Terminal rule

`identify_long_terminals`. For each leaf in each tree:

```
trigger_taxon = (K1 - 1) * median(bl of this taxon across trees) + M_terminal
remove if    bl > trigger_taxon    AND    bl > Q_terminal(q1)
```

A taxon observed in fewer than `MIN_CROSS_GENE_OBS` (= 3) trees is issued no trigger and is therefore never removed by this rule. Its branch lengths still enter the pooled list and the per-unit medians.

3. Named clade stem rule

`identify_clades_with_excessively_long_stems`. Applied per tree, in four steps.

3.1 Resolve each clade to a branch

`resolve_clade_fragments`. Take the clade's taxa present in this tree. If they are monophyletic, that is the fragment. If not, the fragment is the largest monophyletic subset, requiring at least 2 taxa. The measured branch is the one subtending that fragment.

3.2 Collapse aliases

Clade names whose fragments are identical in this tree describe one physical branch and are grouped. Within a group:

Purpose	Which name is used
Which threshold tests the branch	the name with the **smallest** full definition
Which clade is credited in the report	the name with the **largest** full definition
Which clades record the measurement	**all** of them
What enters the pooled background	the branch, **once**

> *The first two point deliberately in opposite directions: the tightest name gives the comparable median, the widest name is what the branch stands for.*

3.3 Test

```
trigger_clade = (K3 - 1) * median(stem of this clade across trees) + M_stem
remove if    stem_bl > trigger_of_tightest_name    AND    stem_bl > Q_stem(q3)
```

A clade whose stem was measurable in fewer than 3 trees gets no trigger and is not testable.

3.4 Suppress nested clades

Removed branches are ordered largest first. A removed branch whose taxa are a subset of one already accepted is dropped from the report: those taxa are already going.

4. Internal branch and long subtree rule

`identify_long_branched_substrees`. Four gates, in this order. All must pass.

Gate 1 — build the chain

```
trigger_tree = (K2 - 1) * median(internal bl in this tree) + M_internal
an internal branch is LONG if   bl > trigger_tree
a STACK is a connected set of long internal branches sharing internal nodes
keep the stack if   length(stack) >= p
```

Built on the trigger alone: the safeguard is deliberately not applied here, so a modest branch in the middle cannot split one chain into two and push both below `-p`.

Gate 2 — safeguard at the branch that acts

`get_outermost_edge` selects from the stack the edge with the most unbalanced split, i.e. the one subtending the smallest clade. That branch is the stem of the clade that would be removed.

```
continue only if   bl_outermost > Q_internal(q2)
```

Gate 3 — side asymmetry

`clade_for_removal_by_asymmetry`. For each side of that branch, the median distance from the branch out to the tips on that side, with the tested branch excluded. This measures how long each side is internally.

```
side_ratio = max(median_u, median_v) / min(median_u, median_v)
continue only if   side_ratio > r
the side to remove is the one with the LARGER median depth
```

Gate 4 — the removed side must be smaller

```
continue only if   length(candidate side) < length(other side)
```

If the deeper side is the larger one, nothing is pruned at that branch. The rule does not fall back to removing the other side.

> ***What this excludes.*** *A small clade of short-branched taxa sitting beyond a long stem fails gate 3: both sides look similar, or the clade side is the shallower one. There the long branch is the stem, not the clade.*

5. Quartet rule

`identify_local_long_branches`. Entirely within one tree; no cross-gene background is used for the trigger.

5.1 Masking, per tree

`detect_mask_groups_for_tree`. For each clade in the `-x` file, keep the largest monophyletic fragment in this tree, minimum 2 taxa. A clade too scattered, or with fewer than 2 taxa present, is not masked in that tree at all.

5.2 Which comparators are allowed

Focal taxon	Comparator	Allowed
not masked	not masked	yes
not masked	in mask group G	no
in mask group G	not masked	yes
in mask group G	same group G	yes
in mask group G	different group G'	no

5.3 Choosing the quartet

Anchors are internal nodes, searched outward from the focal tip level by level, nearest first. At each anchor the branch leading back to the focal tip is dropped and the remaining branches each contribute a group of allowed comparator tips, sorted by distance from the anchor. A tip whose OWN terminal branch is at or below the comparator floor is not a valid comparator: that branch is unestimated rather than short, so it says nothing about the local scale. The test is on the tip's own branch, not on its distance from the anchor - a distance test keeps exactly the tips that should be excluded, because in a cluster of near-identical sequences the accumulated distance exceeds the floor after two or three edges.

```
require at least 2 groups and at least 3 available tips
take the closest tip from each group
if that gives more than 3, keep the 3 closest
if fewer than 3, top up from the pooled remainder, closest first
if 3 cannot be assembled, move to the next anchor
```

The first anchor that yields three comparators is used; the search stops there.

5.4 The test

```
d_focal      = distance(anchor, focal)
local_median = median( distance(anchor, comparator) for the 3 comparators )
ratio        = d_focal / local_median
remove if   ratio > K4   AND   bl_focal > Q_terminal(q1)

not evaluable if   local_median <= comparator floor
```

The safeguard is the **terminal branch-length** quantile, the same value the terminal rule uses, not a quantile of ratios. A trivial branch in a short-branched neighbourhood can produce a huge ratio and is not a long-branch-attraction risk. If no valid quartet can be built, or the local reference still carries no scale, the taxon is retained and counted as not evaluable.

Both guards exist because tree inference pins branches it cannot estimate at a minimum value. Without them, ratios in the thousands appear. Measured on the same dataset at four settings:

Comparator guard	max ratio	ratios > 100	n	99th pct	median
`none`	20805	44	38270	3.19	0.730
`distance from anchor > 1e-6`	20805	54	38270	3.19	0.730
`own branch > 1e-6 (fixed)`	1300	2	38270	3.10	0.728
`own branch > scaled floor`	82.4	0	38270	2.98	0.727

The count n is identical throughout: tightening the guard cost no taxon its quartet, it only replaced comparators that carried no information with ones that did. The fixed 1e-6 was not enough because a branch of 2.54e-6 is not at the floor but is still three orders of magnitude below the real branches around it - it pulled a local median to 7.2e-6 and turned a true ratio of 0.52 into 1280. The Python prototype has no equivalent guard, and never notices because it never pools the ratios.

5.5 One asymmetry between the two scripts

The identifier tests every taxon: a focal tip sitting on an unestimated branch is harmless there, because the terminal-quantile safeguard will retain it anyway. The statistics script additionally **skips** such a focal taxon rather than recording its ratio, since a ratio built on a branch the inference could not estimate would distort the very distribution you are reading K off. This affects the plotted distribution only, never a removal decision.

6. Everything that can block a removal

In every case the outcome is retention.

Rule	Condition	Result
all	unit observed in fewer than 3 trees (terminal, stem)	no trigger issued
all	3 gene trees or fewer in the dataset	terminal, stem and internal rules disabled
all	value exactly equal to a threshold	retained
stem	clade absent, or no monophyletic subset of 2 or more taxa	skipped
stem	taxa already covered by a larger removed clade	not reported again
internal	chain shorter than `-p`	no candidate
internal	outermost branch at or below the internal quantile	rescued
internal	side ratio at or below `-r`	blocked
internal	exact tie between the two side medians	blocked
internal	deeper side is not strictly smaller	blocked
quartet	fewer than 2 groups or fewer than 3 tips at every anchor	not evaluable
quartet	candidate comparator's own terminal branch at or below the comparator floor	not a valid comparator
quartet	local median at or below the comparator floor	not evaluable
quartet (plot only)	focal taxon's own terminal branch at or below the comparator floor	no ratio recorded

7. Parameter map

Flag / file	Position	Feeds
`-t file`	1	K1, terminal trigger
`-t file`	2	K2, internal trigger
`-t file`	3	K3, clade stem trigger
`-t file`	4	K4, quartet ratio threshold
`-s file`	1	q1, terminal quantile - used by the terminal AND quartet rules
`-s file`	2	q2, internal branch quantile
`-s file`	3	q3, clade stem quantile
`-p`	-	minimum branches in a long chain
`-r`	-	side asymmetry ratio, must exceed 1.0

8. Where the code lives

Function	Does
`stats_for_terminals`	per-taxon medians, M_terminal, triggers, pooled list
`stats_for_stems`	as above for clade stems; also collapses aliases per tree
`stats_for_internal_branches`	per-tree medians, M_internal, per-tree triggers
`comparator_branch_threshold`	the comparator floor for this dataset, computed once
`stats_for_quartets`	the ratio distribution (used by the statistics script only)

Function	Does
`resolve_clade_fragments`	clade to fragment and stem length, in one tree
`identify_long_terminals`	rule 1
`identify_clades_with_excessively_long_stems`	rule 2, including alias and nesting
`identify_long_branched_substrees`	rule 3, the four gates
`clade_for_removal_by_asymmetry`	gates 3 and 4
`identify_local_long_branches`	rule 4
`build_quartet_mask_map`, `comparator_allowed`	quartet masking
`quartet_test_for_taxon`	anchor search and the ratio

All of the above are in `Long_Branch_Remover/long_branch_rules.jl`. The libraries TreesIO, TreeUtils and TreeStats hold only general tree operations and no long-branch logic.

> ***One sentence to remember.*** *Each rule asks whether a branch is long relative to its own kind (the trigger, K) and whether it is long in absolute terms across the dataset (the safeguard, q); both must hold, everything ambiguous is retained, and nothing is ever pruned - only listed.*